Extra Exercises 08 – Hallucinations & Fact-Checking

AI Training for Beginners

Self-study training: use AI/LLMs safely in everyday life

Version 1.0 • As of: October 2026

File: `08_extra_exercises.md`

https://github.com/mgunthner/AI-Training

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Contents

Extra Exercises 08 – Hallucinations & Fact-Checking **1**
Task 1 ★ – How thoroughly to check? 1
Task 2 ★★ – The source check 1
Task 3 ★★ – Allowing “I don’t know” 2
Task 4 ★★ – Fact-check plan 2
Task 5 ★★★ – My checking rule 2
Suggested solutions 3

Extra Exercises 08 – Hallucinations & Fact-Checking

Cheatsheet: `01_cheatsheets/08_hallucinations`

Levels: ★ Getting started · ★★ Everyday · ★★★ Challenge **Rule:** Use only invented or anonymized data. All tasks work with any AI service and on a free plan. Suggested solutions are at the end. Only look once you’re done.

Task 1 ★ – How thoroughly to check?

Scenario: You use the AI for eight different things.

Task: Classify each request: **check briefly** or **check thoroughly against the original source**. - a) Three ideas for birthday presents - b) What deadline applies to canceling an online purchase? - c) Making a paragraph of my email friendlier - d) Which dose of a medication is right for me? - e) A quote for my speech - f) A study as evidence for my presentation - g) A summary of a short text I pasted in myself - h) The name of a software extension for my spreadsheet task

Hint: Ask yourself: What happens if the answer is wrong?

Task 2 ★★ – The source check

Scenario: You want to know how reliable your AI’s source references are.

Task: Ask this question and check every source named:

```
What is known about the health benefits of walking?
Name three sources.
```

Fill in the table:

Source	Does it exist?	Does the link open?	Is the claim there?
1			
2			
3			

Hint: If your service has a web search, test both variants (on and off) and compare.

Task 3 ★★ – Allowing “I don’t know”

Scenario: You ask about something the AI can hardly know.

Task: Choose a very small place or a small business near you. Ask in two separate chats:

```
Tell me the history of [small place or business].
```

```
Tell me the history of [small place or business].
If you don't know something for sure, write "I don't know".
Don't make anything up.
```

Compare: How do the answers differ? What can be checked against the place’s local chronicle or website?

Hint: Both outcomes are instructive. Verify at least two claims yourself.

Task 4 ★★ – Fact-check plan

Scenario: The AI wrote this text for your travel guide (all details are placeholders):

```
The museum in Sampletown was founded in year X and is closed on
Tuesdays. Admission costs Y euros for adults. According to a study
by University Z, more than a million guests visit the house each year.
The director, Person A, once said: "Culture is the heart of the city."
```

Task: List every claim. For each claim, note: Do I need to check it? Where do I check it?

Hint: Some details change (opening hours, prices). Others are invented if the source is missing.

Task 5 ★★★ – My checking rule

Scenario: You don’t want to reconsider every time how thoroughly you check.

Task: Fill in the table with your own everyday examples and test it for a week:

Consequences of a mistake	My example	How I check
low		
medium		
high		

Hint: For “high”, the original source or a subject-matter expert always belongs in the mix.

Suggested solutions

Task 1: Check briefly: a, c, g (for g, quickly compare against the original to see whether anything is missing or was made up). Check thoroughly: b (legal matters, deadlines change), d (health: ask a subject-matter expert or check the package insert), e (quotes are readily invented), f (studies can be invented), h (the extension may not exist at all). What matters is your reasoning.

Task 2: The results differ from service to service. Typical: With web search there are more real links; without web search, titles or links are sometimes invented. Even a real source can say something different from what the AI claims. The conclusion: Always open sources yourself.

Task 3: Possible observation: The first answer contains more details, the second admits gaps. If the AI gives many precise details about a very small place, suspicion is warranted. A service with web search may do better here. Verify claims in a reliable source.

Task 4: - Year of founding: yes, official website or chronicle of the museum - Closing day: yes, current website or phone call (changes) - Admission price: yes, current website - Study: yes, does it really exist? Find and read the original - Quote: yes, check wording and source; otherwise don't use it

Task 5: Example: *low* (gift idea): briefly consider whether it fits. *medium* (travel tip): a second source or a website. *high* (deadline, dose, publication): original source or subject-matter expert.

Next: Cheatsheet 09 (Temperature & Sampling), Test B (`03_tests/test_B_good_results`)

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